

QSLAM Highlight Reel — Video Script

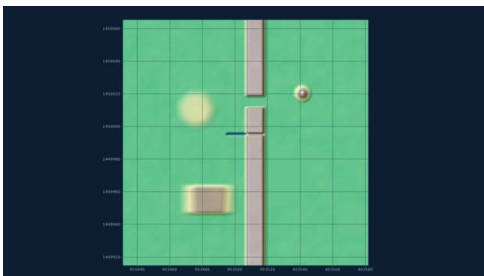
Shot-by-shot content and voiceover for

mission_recordings/report_1401/QSLAM_highlight.mp4 (1920×1080, 30 fps, 1:52.9, silent, no on-screen words)

Does it open with the GeoTIFF + DEM architecture? Yes — the first 19 seconds are the georeferencing chain: GeoTIFF → slope → UTM ↔ map → goal and route. The GeoTIFF *is* the DEM: the elevation data is stored as one georeferenced file, **prior_dem_utm44n.tif**. The reel skips two scenes from **geo_pipeline.mp4**: the DEM height-profile sweep (0:08–0:18 in that file) and the occupancy-grid fade (0:28–0:35). Cut those in from there if you want “DEM” shown on its own.

The video carries no words or audio. The VO lines below are for you to speak over it, sized at about 2.5 words per second. Timings are checked against the file (112.93 s) and against frames taken every 2 s.

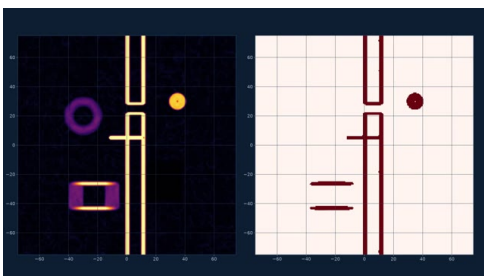
Part 1 — The input: GeoTIFF → goal (0:00 – 0:19)



0:00 – 0:04 • GeoTIFF

On screen: The elevation raster (green tint) is revealed north to south on its own UTM zone 44N grid — Eastings 403440–403580, Northings 1449920–1450060. The berm (vertical wall), a mound, a dome and a raised block are visible.

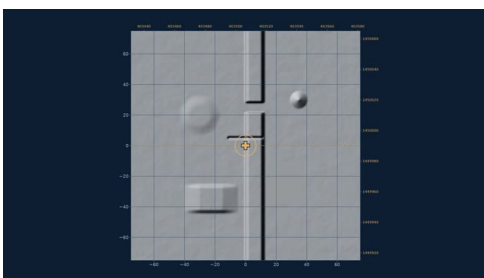
VO: “The mission starts from one file: a GeoTIFF elevation model in UTM zone 44 North.”



0:04 – 0:09 • Slope → blocked cells

On screen: Left: slope computed from the DEM — berm edges, the mound’s ring and the dome glow. Right: cells turn dark red (blocked) as the slope threshold sweeps down to the rover’s 15° limit.

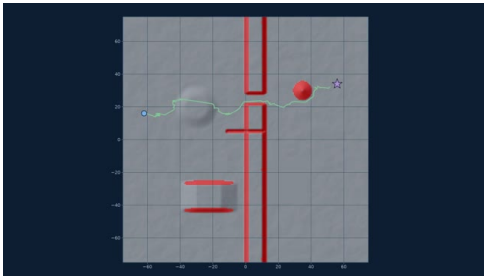
VO: “We compute slope from the DEM. Anything steeper than fifteen degrees is marked as blocked.”



0:09 – 0:13 • UTM ↔ map frame

On screen: Grey hillshade with map-frame metres on the left and bottom, and UTM numbers in gold on the top and right. Dashed gold crosshairs draw in; the surveyed anchor appears at (0, 0) with a ring.

VO: “A surveyed anchor ties UTM to the rover’s map frame. That one point links the two coordinate systems.”

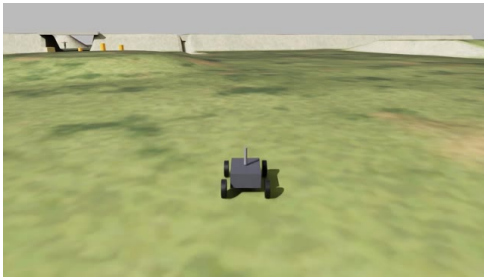


0:13 – 0:19 • Goal and route

On screen: Blocked terrain in red. Blue dot = start; a violet star appears at the goal. A green line traces the route actually driven: east across the field, through the tunnel in the berm, around the dome, to the star.

VO: “The goal is given as a UTM grid reference, E 403562, N 1450024, which converts to map point 56, 34. This is the route the rover drove.”

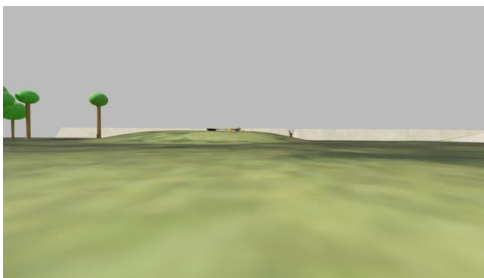
Part 2 — The drive (0:19 – 1:05)



0:19 – 0:27 • Chase camera, first motion ($\approx 2.5\times$ speed)

On screen: Third-person view from behind as the rover pulls away across grassy terrain, with the berm on the horizon.

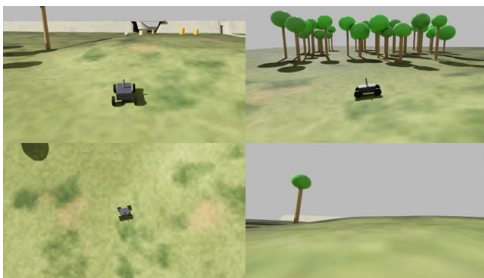
VO: “No GPS is used. The rover starts moving using only its wheels, IMU and lidar.”



0:27 – 0:35 • Rover's own camera ($\approx 3\times$ speed)

On screen: The mast-mounted ZED camera view: open ground, rolling terrain, a line of trees, and the berm ahead.

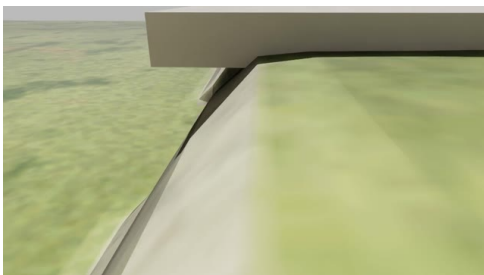
VO: “This is what the rover sees from its own camera.”



0:35 – 0:51 • 2x2 camera grid — the long traverse ($\approx 26\times$ speed)

On screen: Chase, side, top-down and onboard views together. The rover weaves past trees and yellow barrels as it closes in on the berm.

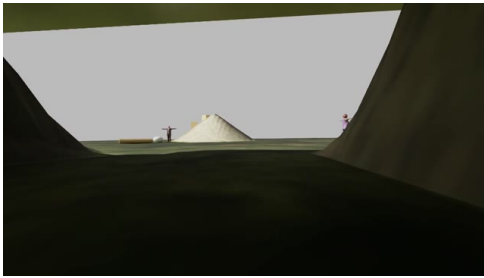
VO: “Seven minutes of driving in sixteen seconds. The rover avoids trees and barrels as it goes, and the planner replans as the lidar picks up new obstacles.”



0:51 – 0:57 • Side camera — berm approach ($5\times$ speed)

On screen: The rover runs along the 3.2 m berm wall, passes an orange barrel, and lines up with the tunnel entrance.

VO: “The berm is too steep to climb, so the only way through is the tunnel.”

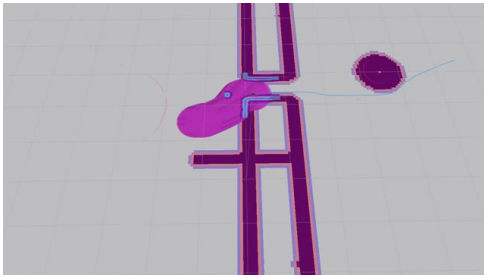


0:57 - 1:05 • Onboard camera — through the tunnel ($\approx 4\times$ speed)

On screen: The berm wall fills the frame, the view goes dark inside the tunnel, then the exit opens up with a mound and a standing figure beyond it.

VO: “Inside the tunnel. The lidar keeps the pose estimate going with no external position fix.”

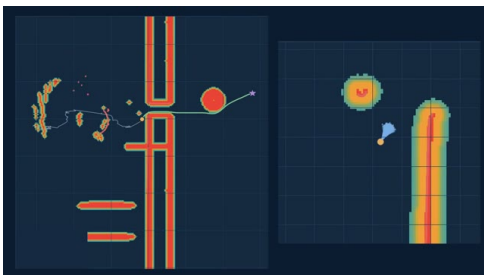
Part 3 — Inside the stack (1:05 - 1:44)



1:05 - 1:14 • RViz — the operator’s view

On screen: 3D RViz view: the prior map’s berm and dome in purple, the inflation zone in magenta around the rover at the tunnel mouth, and a blue planned path curving around the dome.

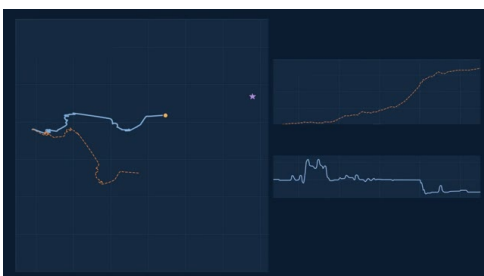
VO: “This is the operator’s live view: the prior map, the costmap and the planned path.”



1:14 - 1:25 • Costmaps and planning (whole mission, $\approx 77\times$)

On screen: Left: global costmap — red lethal cells with an inflation border from the GeoTIFF slope prior; lidar hits (tree clusters) are added as the rover drives; the Smac A* route runs to the star. Right: rolling local costmap with a fan of MPPI candidate paths ahead of the rover.

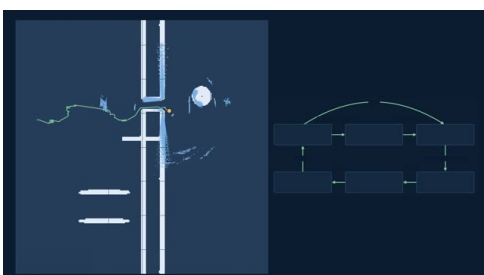
VO: “The global plan comes from the GeoTIFF prior plus live lidar, using Smac A*. MPPI handles local control.”



1:25 - 1:35 • EKF state estimation ($\approx 84\times$)

On screen: Three tracks: ground truth (thick grey), EKF estimate (blue, lying on top of it) and wheel-only dead reckoning (orange dashed, drifting away). Upper-right graph: dead-reckoning error climbing. Lower-right graph: heading over time.

VO: “Wheel odometry alone drifts badly. The fused EKF stays on the true path: 0.33 m RMS error, 0.73 m at arrival.”

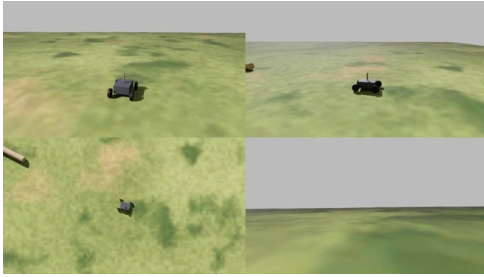


1:35 - 1:44 • SLAM map and ROS 2 graph ($\approx 94\times$)

On screen: Left: the lidar map builds up in blue-white over the prior, with the trajectory and pose-graph nodes. Right: the ROS 2 data-flow loop of six boxes (unlabelled in the reel): Isaac Sim → EKF → slam_toolbox / map_server → costmap → Smac A* → MPPI. Each link lights up while its topic carries data.

VO: “slam_toolbox builds the map as the rover drives. The full loop runs live: sensors, EKF, map, costmap, planner, controller.”

Part 4 — Arrival (1:44 – 1:53)



1:44 – 1:53 · 2×2 camera grid — final approach

(≈5× speed)

On screen: All four cameras follow the rover over the last stretch of ground as it slows and stops at the goal.

VO: “Goal reached. 187.5 metres driven, stopped 17 centimetres from the target, with no GNSS used at any point.”

Numbers used in the VO

From this run’s mission_log.csv and VIDEOS.md.

Commanded goal	UTM 44N E 403562.52 N 1450024.12 → map (56.00, 34.00)
Anchor (map origin)	E 403506.52 N 1449990.12
Outcome	Goal reached (one recovery behaviour triggered)
Distance driven	187.5 m
Final distance to goal	0.17 m
EKF error	0.73 m at arrival · 0.33 m RMS
Mission length	≈14 min sim time (847.6 s)
GNSS	No GNSS receiver read at any point

Timing note: 0:00–0:19 has four quick cuts. Short lines fit, but the goal-coordinate line (0:13–0:19) is tight. If it feels rushed, drop “which converts to map point 56, 34”.